

MATHEMATICS DEPARTMENT, UNIVERSITY OF MASSACHUSETTS DARTMOUTH
Scientific Machine Learning
MTH 602 — Fall 2026
Course Information and Syllabus
Updated August 31, 2026

Instructor: Scott Field
Instructor email: sfield@umassd.edu
Office: NA (Spruce, probably 172, 175, 177, or 178 as available)
Office hours: See the course website for time & location
Class time and location: Monday/Wednesday, 11:00AM-12:15PM Textiles (TEX) 001

Course Websites There are two online resources associated with this course

Main: <https://github.com/sfield17/MTH602-Fall2026-SciML>
Discussion: Slack Unity User Community (channel: `sciml-mth602-fall2026`)

Prerequisites This course will be mixing ideas from high performance computing, numerical analysis, computational mathematics, and machine learning. I will assume minimal knowledge of machine learning, but I will assume you are comfortable with the topics covered in EAS 520/DSC 520, EAS 501, and EAS 502. These classes cover topics like multivariable calculus, numerical linear algebra, solving ODEs and PDEs (numerically), quadrature, Fourier series, operators, programming, running code on remote clusters, and using git. Concepts like Runge-Kutta methods, explicit vs implicit solving, etc should be familiar too.

Textbooks and other resources We will be using the book by Bishop and Bishop, although we won't follow the book throughout the entire semester. We will frequently use open-source supplemental materials and lecture notes as well.

1. (Required) Christopher M. Bishop and Hugh Bishop, *Deep Learning: Foundations and Concepts*
2. (Optional) William H. Press, Saul A. Teukolsky, William T. Vetterling, Brian P. Flannery *Numerical Recipes: The Art of Scientific Computing*
3. (Optional) Alfio Quarteroni, Riccardo Sacco, and Fausto Saleri *Numerical Mathematics*

Course description The course catalog says “Scientific machine learning algorithms for computational science and engineering. Topics may include physics-informed neural networks, neural dynamical systems, AI-based surrogate models, signal detection with convolutional neural networks, learning nonlinear continuous operators, neural turbulence models, optimization algorithms, simulation-based Bayesian inference, and more. Python will be the primary language. Emphasis on real-world applications, covering high-performance computing with multi-core and GPU acceleration.”

As you can see from the catalog's description, MTH 602 is a special topics course for Scientific Machine learning. This semester we begin with foundational deep learning topics as covered in the first few chapters of the book *Deep Learning: Foundations and Concepts*. The second half of the semester will cover SciML topics such as function approximation (AI-based models), signal detection with convolutional neural networks, learning dynamical systems, physics-constrained optimization problems, distribution sampling, and PINNS. Time permitting, more topics will be covered based on the interest of the class.

The material will be motivated by solving challenging scientific problems. As such, there will be a mixture of traditional paper-and-pencil problems as well as writing and running code on computing resources. Throughout the course we will often explore these topics on our regional supercomputer Unity, a million dollar GPU-based system hosted at the MGHPCC (<https://docs.unity.rc.umass.edu/>).

Learning outcomes At the end of the semester, you will be able to

- apply scientific machine learning techniques to traditional scientific tasks
- assess the benefits and drawbacks of scientific machine learning techniques when compared to traditional methods
- understand and implement state-of-the-art scientific machine learning algorithms from recent journal articles
- generalize and apply concepts from this class to areas of interest to you

Grading Policy Your grade will be determined by homeworks (20%), exam I (15%), exam II (15%), exam III (20%), additional activities (10%), and a final project (20%). When computing your grade, please keep in mind how rounding will be carried out: after a grade computation I will apply standard rounding rules (e.g. $80.5 \rightarrow 81$ and $80.4999999 \rightarrow 80$).

A	93% - 100%	B-	80% - 82%	D+	67% - 69%
A-	90% - 92%	C+	77% - 79%	D	63% - 66%
B+	87% - 89%	C	73% - 76%	D-	60% - 62%
B	83% - 86%	C-	70% - 72%	F	0% - 59%

A grade of an A+ will be given for exceptional performance.

Homework policy Generally you will have two weeks to complete each homework project. Expect about 7 or so homeworks throughout the semester.

You may discuss homework questions with other students in the class (please ask questions on Slack!), **but you must individually write your own code and solutions as a group.** I will post homework solutions on the course website once I have received all the assignments. Plagiarism from any source is not acceptable and will result in a failing grade for the assignment or, in severe cases, a failing grade for the course in addition to any University penalties.

I will post homework solutions once I have received all the assignments.

Plagiarism and using AI To summarize key points:

- You may discuss homework questions with other students in the class and some projects will be group projects.
- You can use the internet, AI tools, and other students to help you understand how to write code
- You cannot directly copy code from other groups/students and AI and use it as your own
 - Using copilot is OK. Ask AI for help as you would ask me for help is ok
 - Asking AI to solve the entire problem for you is Plagiarism.
 - **Working rule:** if you wouldn't ask me to do it, don't ask the AI.
- Any code you use from the internet should be cited (e.g. from Stack Overflow)
- You must understand what your doing – 50% of your grade is from exams. The exams are be designed to be straightforward if you did the homework.

Homework solutions Your homework solution should consist of a single, well-formatted document that includes the results of your work (figures, plots, tables, descriptions, mathematical expressions, etc.) and any relevant/interesting portions of your computer code used to generate the results. You are encouraged to write your solution using either L^AT_EX or a Jupyter notebook, but any reasonable document preparation software is OK. **Only PDF format will be accepted.**

Most homeworks will require writing and running computer code. Your code should be tracked by the version control system git. Students are responsible for setting up their own git repo, pushing their work their, and sharing the git repo information with the instructor. I should be able to run your code on my laptop and reproduce results described in your report. You should aim to write high-quality code which is easily readable by others and clearly commented. Code which is impossible to read or does not run will result in point deductions.

Some problems will be of the “paper-and-pencil” variety. You can either write this up as part of your solution document or turn in handwritten work.

Submitting homework Your homework will always be due on a class day. You should strive to turn in your homework at the beginning of class. If this is not possible, you must find some other way to get me your homework before midnight on the day it is due.

Final project Your final project will be designed and carried out by you. It should relate to your ongoing MS or PhD research projects.

What are “additional activities”? These will come in different forms, but will largely consist of (i) leading discussions on a sciML topic, (ii) discussing homework solutions, (iii) any in-class problem solving as they arise, and (iv) short computational lab assignments. These will be graded as “check plus” (all work completed with good effort), “check” (missing work, minimal effort), and “No Credit” (nothing was done). These activities will be described as they arise and listed on the class website.

Computing lab Some classes will be devoted to working on short assignments, running tutorials, or run as a recitation-style homework session. This work will require a computer, ideally your laptop. Please bring a laptop to every class.

Programming languages The homeworks will require you to do some computer programming. You may use any (reasonable) programming language you would like, especially if you are already using this language in your own research. I would strongly encourage students to use either Python or Julia. The class will be taught in Python. *Please speak with the instructor if you are planning to use a language other than Python or Julia.*

Class structure You should take notes in class, I may not post lecture notes. I will post all classroom demonstrations and PowerPoint presentations. Please, no distracting items (phones, ipods, etc) in class.

Class communication An email list will be used to send out information not communicated during class or issues affecting the the logistics of the class. If you are registered for the course, but do not receive the course email announcements, please notify me immediately. The course website will also be used to communicate more technical matter of the class (e.g. homework assignments) and should be regularly checked.

Attendance I do not take attendance; attendance at class meetings is not a factor in your grade. However, you are responsible for your own awareness of any course material, policies and practices that are discussed in class. Since part of the class meeting time will be devoted to homeworks and labs, there is little reason to not attend.

If you believe you will be absent for an extended period of time due to illness or other emergency, notify me immediately to arrange make-up of any assignments that will be missed. No assignments can be “made up” after an absence, except in extenuating circumstances.

Incomplete grades An incomplete grade will be given only in exceptional circumstances at my discretion. You must be passing the class at the time of the request for an incomplete grade, or be sufficiently close to passing. If the work is not completed within one year of the recording of the incomplete grade, the grade will become an F(I). The incomplete policy for this course is that at least 70% of the course must be already completed and an exceptional circumstance (i.e. medical issue) must exist. If you feel you require an incomplete for an exceptional reason, you need to email me and state your reasons for the incomplete in writing. We will then schedule a meeting and decide how to proceed.

Students with disabilities If you have a documented disability and require accommodations to obtain equal access in this course, you will need to self-disclose by calling (508.999.8711) or emailing (access_success@umassd.edu) the Office of Student Accessibility Services to make an appointment with one of the directors. Returning students need to register each semester.

Academic integrity policy Our class will follow UMassD’s academic integrity policy that can be found here <https://www.umassd.edu/studentaffairs/studenthandbook/academic-regulations-and-procedures/>.

Additional University-wide policies https://www.umassd.edu/media/umassdartmouth/provost/omnibus_language_for_syllabi_-jan_11_2019.pdf

Important dates

Wednesday, September 2, 2026	Classes begin
Monday, September 7, 2026	No classes
Monday, September 14, 2026	Last day to add, drop, or audit
Thursday, October 8, 2026	Last day to request pass/fail
Monday, October 12, 2026	No classes
Tuesday, October 13, 2026	Class!!!
Monday, October 19, 2026	Exam I
Wednesday, October 28, 2026	No classes
Friday, November 13, 2026	Last day to withdraw from a class
Wednesday, November 18, 2026	Exam II
Wednesday, December 9, 2026	Classes end
Wednesday, December 9, 2026	Final Presentations
Thursday, December 17, 2026	Exam III (11:30am-2:30pm)